

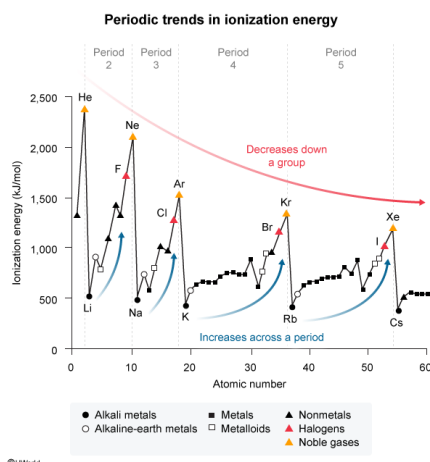
Which of the following reactions requires the smallest amount of energy to remove a valence electron (e^-)?

- ☐ A. $F(g) + \text{energy} \rightarrow F^+(g) + e^-$ [8%]
- ☐ B. $Ne(g) + \text{energy} \rightarrow Ne^+(g) + e^-$ [3%]
- ✓ ☒ C. $Cl(g) + \text{energy} \rightarrow Cl^+(g) + e^-$ [81%]
- ☐ D. $Ar(g) + \text{energy} \rightarrow Ar^+(g) + e^-$ [5%]

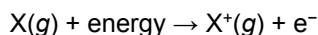
Correct

81% answered correctly

Explanation:



The **first ionization energy** is the energy required to **remove the first**, most loosely bound **valence electron** (e^-) from a single neutral atom (X) in the gaseous state. The process can be represented generally as:



Elements with loosely bound valence electrons have lower first ionization energies (ie, are easier to ionize) than elements with tightly bound valence electrons. On the periodic table, the **first ionization energy** tends to **increase** with increasing atomic number moving **across a period** (row), but it tends to **decrease** moving **down a group** (column), with some intermittent exceptions.

In this question, each of the given reactions describes the first ionization of an element. Ne and Ar are both noble gases (Group 18) and are further across their respective periods than the halogens F and Cl (Group 17). As a result, each halogen has a lower first ionization energy than its neighboring noble gas on the periodic table.

Comparing the halogens, the atoms in the group **increase in size** moving down the column, which causes the valence electrons in larger halogens to occupy a **higher energy level** where they are less tightly bound. As a result, removing valence electrons from larger halogens requires less energy than removing valence electrons from smaller halogens.

Because halogens have a lower first ionization energy than their neighboring noble gases, and because Cl is larger and farther down the halogen group than F, the first ionization of Cl will require the smallest amount of energy of the given reactions.

(Choices A, B, and D) Noble gases such as Ne and Ar have a *higher* first ionization energy within their respective periods. Cl is larger in size and lower in the halogen column than F, giving F a *higher* first ionization energy than Cl.

Educational objective:

The first ionization energy is the energy required to remove the first, most loosely bound valence electron from a neutral atom. The first ionization energy tends to increase with increasing atomic number moving across a period and decrease moving down a group on the periodic table.

Time spent: 0 sec
Subject:
General Chemistry

QID: 402515
System:
Atomic Theory and Chemical Composition

The effective nuclear charge experienced by the valence electrons of a neutral nitrogen atom is:

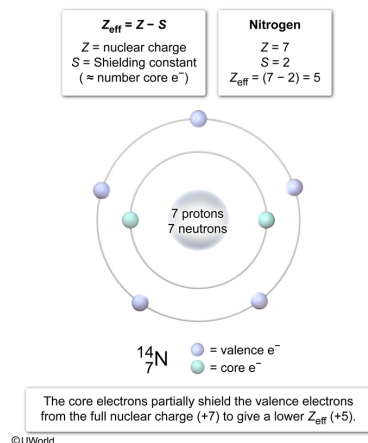
- ☐ A. +7. [11%]
☒ B. +5. [54%]
☐ C. -3. [23%]
☐ D. -5. [10%]

Correct

54% answered correctly

Explanation:

Effective nuclear charge Z_{eff} of nitrogen



Each proton in the nucleus of an atom contributes one unit of positive charge that exerts an **electrostatic attraction** on the negatively charged electrons around the atom. However, in atoms with several electrons, the **core electrons** positioned between the valence electrons and the nucleus provide a **shielding constant S** that counteracts part of the full **nuclear charge Z** attracting the valence electrons. As a result, the valence electrons experience an **effective nuclear charge Z_{eff}** that is less than the full nuclear charge, such that

$$Z_{\text{eff}} = Z - S$$

Precisely calculating S is difficult and depends on the electron density of a given electron orbital configuration, but a rough, first-order approximation can be made by assuming that S is approximately equal to the number of core electrons.

When comparing atoms within the same period (row) of the periodic table, this approximation shows that Z_{eff} increases as the **atomic number** increases. However, because this approximation does not account for electron density or orbital distance, it is less useful for precisely comparing Z_{eff} between atoms in the same group (column) or in different rows of the periodic table.

Applying this approximation to calculate the Z_{eff} experienced by the valence electrons of a nitrogen atom with 7 protons ($Z = 7$) and 2 core electrons ($S = 2$) gives:

$$Z_{\text{eff}}, \text{ N} = Z -$$

(Choice A) A charge of +7 is the full nuclear charge of the nitrogen atom nucleus rather than the Z_{eff} experienced by the valence electrons.

(Choice C) A net ionic charge of -3 is obtained by a nitrogen atom that gains three additional electrons, but Z_{eff} is the amount of charge from the *nucleus* influencing the valence electrons.

(Choice D) The charge from the 5 valence electrons in a nitrogen atom is -5, but Z_{eff} is the amount of the positive charge from the nucleus that is experienced by the valence electrons.

Educational objective:

The nuclear charge Z influencing a valence electron is diminished by the shielding constant S , which results in an effective nuclear charge Z_{eff} that is less than Z and increases as the atomic number increases. By assuming S is equal to the number of core electrons, Z_{eff} can be estimated as: $Z_{\text{eff}} = Z - S$.

Time spent: 0 sec
 Subject:
 General Chemistry

QID: 402516
 System:
 Atomic Theory and Chemical Composition

A sample of potassium-42 was measured to have the radioactivity shown in the table below.

Time (h)	Activity (Bq)
3	3112
6	2616
9	2200
12	1850
15	1556

Based on the given data, what must the initial activity of the sample have been at zero hours?

- ☐ A. 6223 Bq [3%]
- ☐ B. 5232 Bq [3%]
- ☐ C. 4400 Bq [8%]
- ✓ ☒ D. 3700 Bq [84%]

Correct

83% answered correctly

Explanation:

Determining half-life and initial activity of potassium-42

half-life = 12 hours

Time (h)	Activity (Bq)
3	3112
6	2616
9	2200
12	1850
15	1556

12 hours pass

Activity drops by half

Time (h)	Activity (Bq)
0	3700
3	3112
6	2616
9	2200
12	1850
15	1556

12 hours earlier

Initial activity 2 times higher

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In **radioactive decay**, the **half-life** of a radioisotope is the time required for the amount of the isotope to **decrease by half** (50% of the initial amount). With each subsequent half-life interval that passes, the amount of the isotope that remains is again decreased by half so that the fraction of the initial amount remaining becomes **exponentially smaller** and eventually approaches zero. Accordingly, the fraction remaining after n half-lives can be calculated as:

$$\text{Fraction remaining} = \frac{1}{2^n}$$

The activity of a radioisotope refers to the number of decay events per unit of time. In the International System of units (SI), the preferred unit of activity is the becquerel

(Bq), in which 1 Bq equals 1 radioactive decay per second. Accordingly, radioisotope activity is proportional to the amount of the decaying radioisotope present, and activity decreases by 50% with each successive half-life that passes.

$$\text{Measured activity} = (\text{Fraction remaining}) \times (\text{Initial activity})$$

$$\text{Measured activity} = \frac{1}{2^n} \times (\text{Initial activity})$$

In this question, the half-life is not directly given and must be determined from the data table. This can be done by selecting a convenient starting time and then seeing how much additional time must pass for the activity to drop by half from that point. If the activity after 3 hours is chosen, the additional time required to decrease the amount of ^{42}K by 50% (the half-life) is 12 hours.

$$3112 \text{ Bq at 3 hours} + 12 \text{ hours} \implies \left(\frac{1}{2} \times 3112 \text{ Bq} = 1556 \text{ Bq} \right) \text{ at 15 hours}$$

This means that after 12 hours (one half-life, $n = 1$) the measured activity (1850 Bq in this case) is only half of the initial activity that was present at time zero. Therefore, the initial activity of the ^{42}K sample must have been two times higher.

$$\text{Initial activity} = 2^n \times (\text{Measured activity})$$

$$\text{Initial activity} = 2 \times (1850 \text{ Bq}) = 3700 \text{ Bq}$$

(Choices A, B, and C) Based on a half-life of 12 hours, an initial activity greater than 3700 Bq would be too high to result in the measured activities in the table during the time intervals listed. Times prior to time zero would be negative and conceptually invalid.

Educational objective:

The activity of a radioisotope (the number of decay events per unit of time) is proportional to the amount of the decaying radioisotope present and decreases by 50% for each radioisotope half-life that passes. With each subsequent half-life, the initial amount of isotope that remains becomes exponentially smaller and eventually approaches zero.

Time spent: 0 sec

Subject:

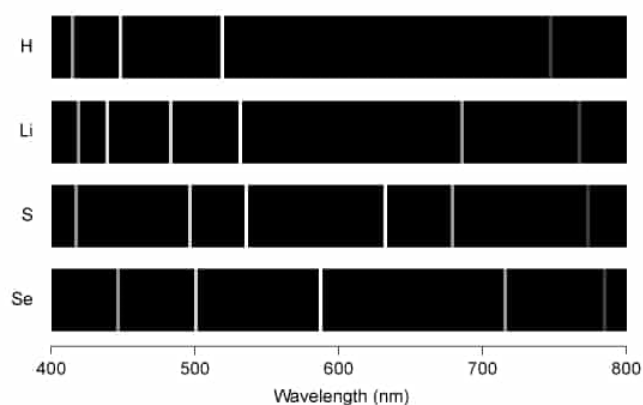
General Chemistry

QID: 402517

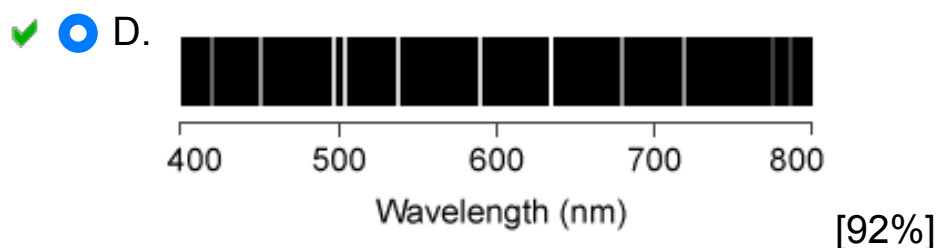
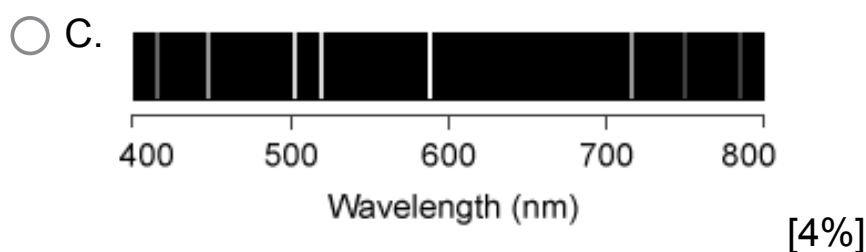
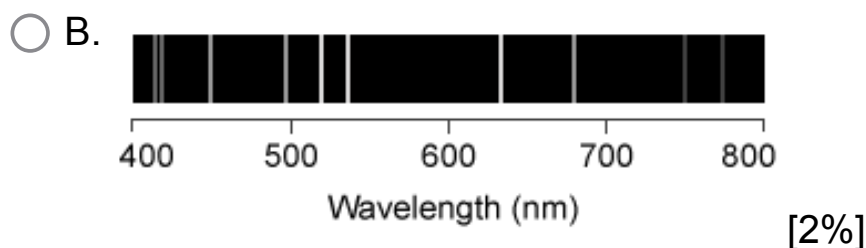
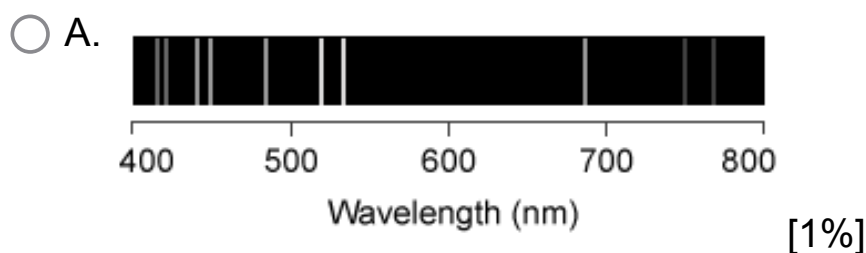
System:

Atomic Theory and Chemical Composition

The emission line spectra of hydrogen, lithium, sulfur, and selenium for the visible region of the spectrum are shown below.



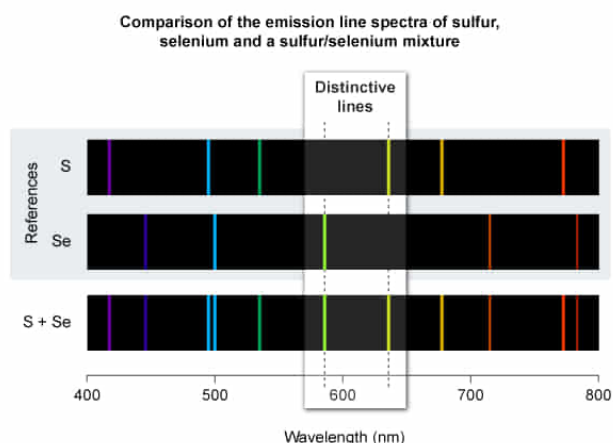
If a sample contains more than one of these elements, which of the following spectra would correspond to a sample containing both S and Se?



Correct

91% answered correctly

Explanation:



Electrons within an atom can exist only in particular, discrete energy levels that are identified by **quantum numbers** (such as the principle quantum number n) and by **orbital** designations. An **electron excited** from a lower to a higher energy level must absorb an amount of energy equal to the energy difference between the two levels, but an electron moving from a higher to a lower energy level releases energy (in the form of a photon) with an energy equal to the energy difference between the levels.

Because **each element** has a unique **electron configuration**, excited electrons in a gas-phase sample of a pure element **emit photons** only at **specific wavelengths** rather than as a continuous spectrum. This produces a **line spectrum** with emission lines at particular wavelengths that are characteristic of the given element and its electron configuration. As a result, the **line spectrum** of each element is **unique** and can be used like a fingerprint to identify elements.

As seen from the reference spectra given for this question, each element has a different spectral pattern. Moreover, S and Se also have distinctive lines in the range of 575-650 nm, which are absent in the spectra of Li and H. Specifically, Se displays a line near 585 nm, whereas S displays a line near 630 nm. As such, the line spectrum of a sample mixture that contains both S and Se must have the spectral patterns of S and Se, which includes lines at both 585 and 630 nm. Of the four choices, only **(Choice D)** is a match.

(Choice A) This is the line spectrum of a mixture of **H and Li**, which lacks the

lines indicative of S and Se.

(Choice B) This is the line spectrum of a mixture of H and S, which lacks the lines indicative of Se.

(Choice C) This is the line spectrum of a mixture of H and Se, which lacks the lines indicative of S.

Educational objective:

Each element emits photons only at specific wavelengths to give a unique emission line spectrum corresponding to its electron configuration.

Time spent:0 sec

QID:402518

Subject:
General Chemistry

System:
Atomic Theory and Chemical Composition